

Claim	Checked how	Result
The final bundle is complete and internally attributable	Notebook 02 verifies all 106 manifest hashes; 43 result tables and the publication-completeness gates are present	Confirmed
56,194 windows use a 45,784/7,950/2,460 chronological split	Reconciled saved window metadata, split labels, and the 59,040 horizon-level test predictions	Confirmed
Ridge is the strongest tested forecaster	Recomputed all model metrics: C01 Ridge has MSE 39.144 and MAE 4.079; M13 TRACE has MSE 40.062 and MAE 4.134	Corrected
TRACE’s mechanism ladder is measured, not assumed	A04 raw fusion selects weight 0; A05 ungated residual has MSE 44.959; A06 constant residual selects strength 0; M13 selects gate strength 0.25	Confirmed
TRACE significantly improves its C06 base ensemble	M13–C06 MAE difference is -0.0212 ; unadjusted 95% CI $[-0.0383, -0.0046]$, but Holm-adjusted bootstrap $p = 0.355$ and DM $p = 0.381$	Corrected
The event channel provides a demonstrated improvement	Compared M13 with event-free A03: MAE differs by -0.0003 across only 70 event-context test origins; adjusted $p = 1.000$	Flagged unverified
Output-space retrieval improves the tested frozen Chronos-Bolt configuration	M11 worsens MAE by 0.3718 versus M10; Holm-adjusted bootstrap $p = 0.046$ and DM $p = 0.035$	Corrected
The final forecast is exactly decomposable	Verified pointwise that M13 equals base forecast plus gate times residual; mean effective gate is 0.090 and mean absolute correction is 0.149	Confirmed
NOAA Storm Events supports strict operational availability	Audited source fields: event start is used as availability because machine-readable issue timestamps are absent	Flagged unverified
The shift score establishes general concept-drift adaptation	The implemented diagnostic is reproducible, but detector agreement is 11.17% and no causal change in the predictive relationship is established	Corrected
Findings generalize geographically and runtime is hardware-normalized	The final evidence covers one Los Angeles County aggregate; no external county or hardware-normalized benchmark is present	Flagged unverified

AI writing removed or corrected (before → after).

Foundation model	“TRACE-RAF is a foundation model; foundation-model retrieval is unsupported”	TRACE-RAF is a supervised residual-correction model; only event-conditioned output fusion with the tested frozen Chronos-Bolt configuration is unsupported here
Event awareness	“Events improve forecasting”	The retrospective event channel is investigated; this holdout does not establish an event-specific gain
Superiority	“TRACE beats the baselines”	Ridge is best numerically; TRACE is competitive, but its base-model gain is not significant after family-wise correction
Shift terminology	“Concept-drift detector”	Distribution-shift diagnostic and model-specific regime label, without a causal concept-drift claim
Explainability	“Causal explanation”	Auditable evidence records and an exact forecast decomposition into base, residual, and gate terms